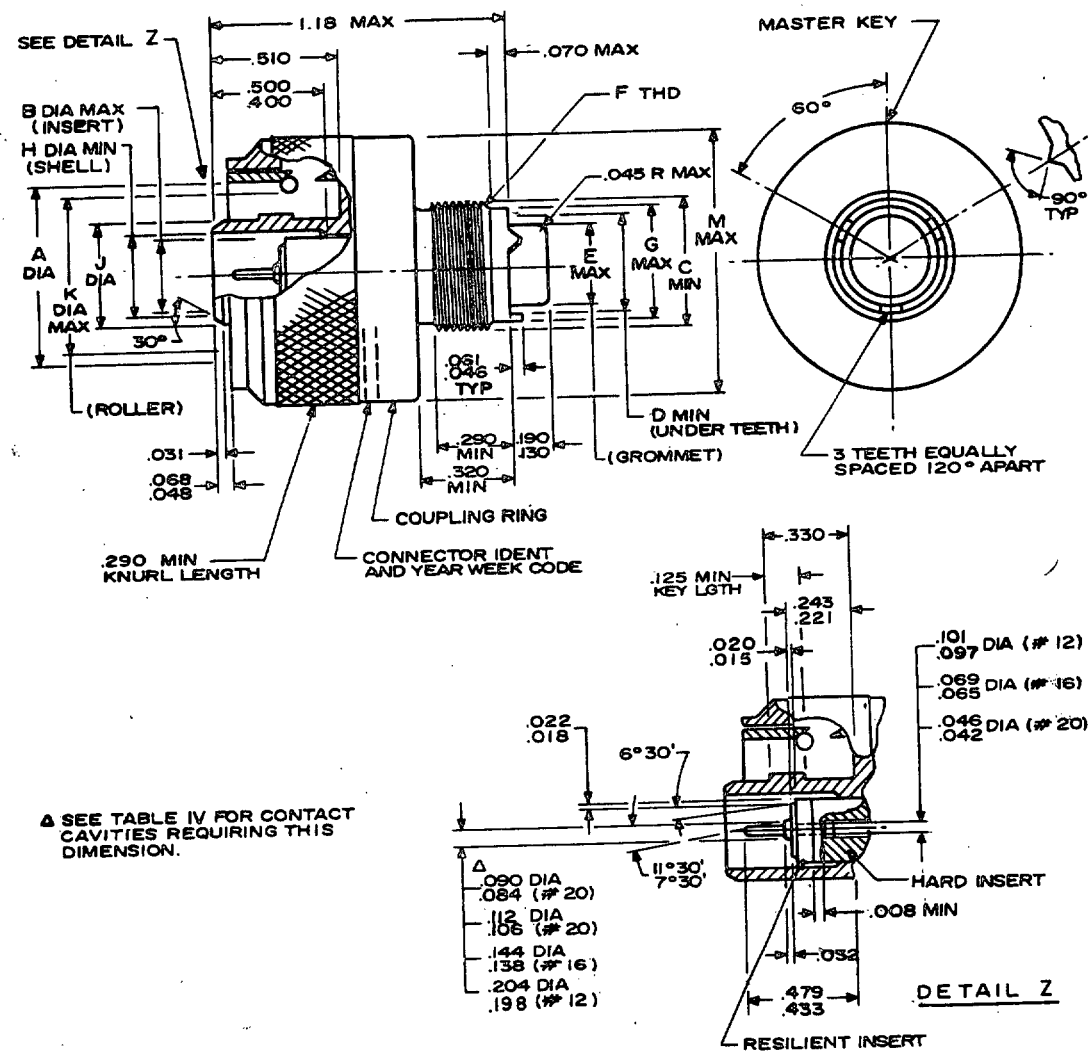
MILITARY SPECIFICATION

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING) PLUG, THREADED COUPLING, (PUSH-PULL, QUICK-DISCONNECT WITHOUT LANYARD), CRIMP PIN CONTACT, SERIES III, CLASSES A, G, AND R

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-83723(USAF).



Δ SEE TABLE IV FOR CONTACT CAVITIES REQUIRING THIS DIMENSION.

FIGURE 1

FSC 5935

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TABLE I

SHELL SIZE	A	B	C	D	E	F THREAD	G	H	J	K	L	M	N
8	.704	.290	.470	.370	.305	.500-20 UNF-2A	.437	.294	.405 .400	.612	.087 .081	1.095	.225 .222
10	.829	.388	.600	.497	.405	.625-24 UNEF-2A	.572	.428	.526 .521	.737	↕	1.240	.285 .282
12	1.016	.558	.724	.613	.531	.750-20 UNEF-2A	.687	.598	.696 .691	.925	↕	1.432	.370 .367
14	1.079	.627	.849	.738	.665	.875-20 UNEF-2A	.812	.667	.765 .760	.987	.087 .081	1.490	.405 .402
16	1.204	.772	.974	.863	.790	1.000-20 UNEF-2A	.937	.794	.992 .887	1.112	.118 .112	1.711	.468 .465
18	1.345	.860	1.030	.919	.869	1.0625-18 UNEF-2A	.992	.900	.998 .993	1.237	↕	1.815	.521 .518
20	1.470	.985	1.154	1.044	.994	1.1875-18 UNEF-2A	1.117	1.025	1.123 1.118	1.362	↕	1.962	.583 .580
22	1.595	1.110	1.279	1.169	1.119	1.3125-18 UNEF-2A	1.242	1.150	1.248 1.243	1.487	↕	2.070	.646 .643
24	1.720	1.235	1.404	1.294	1.244	1.4375-18 UNEF-2A	1.367	1.275	1.373 1.368	1.612	.118 .112	2.195	.708 .705

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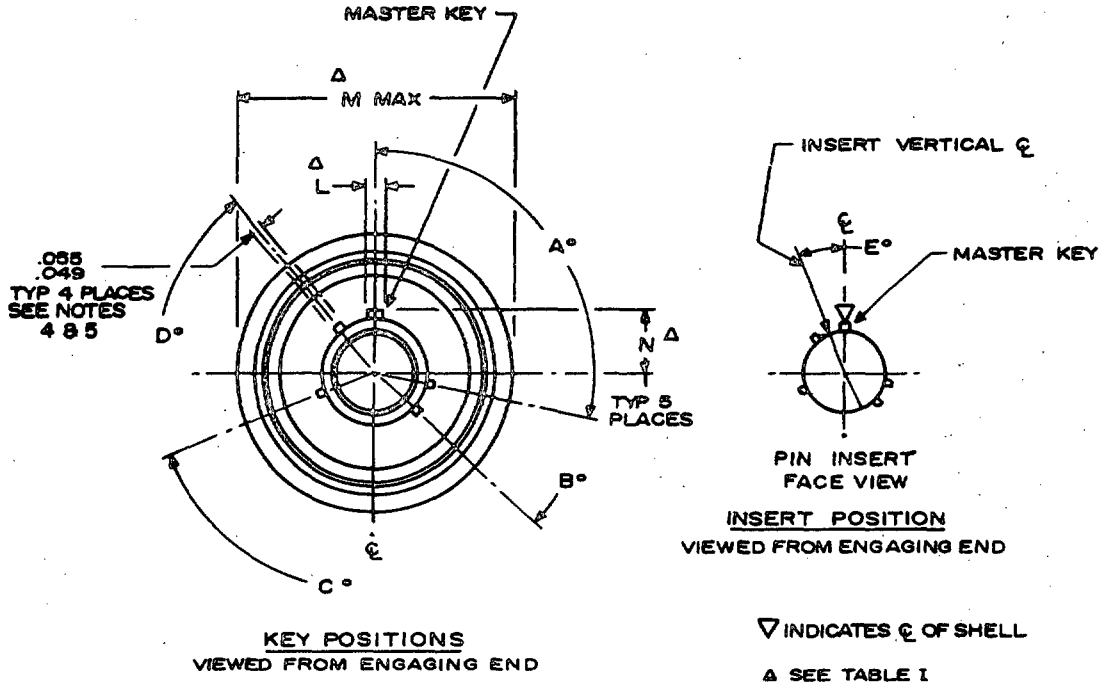
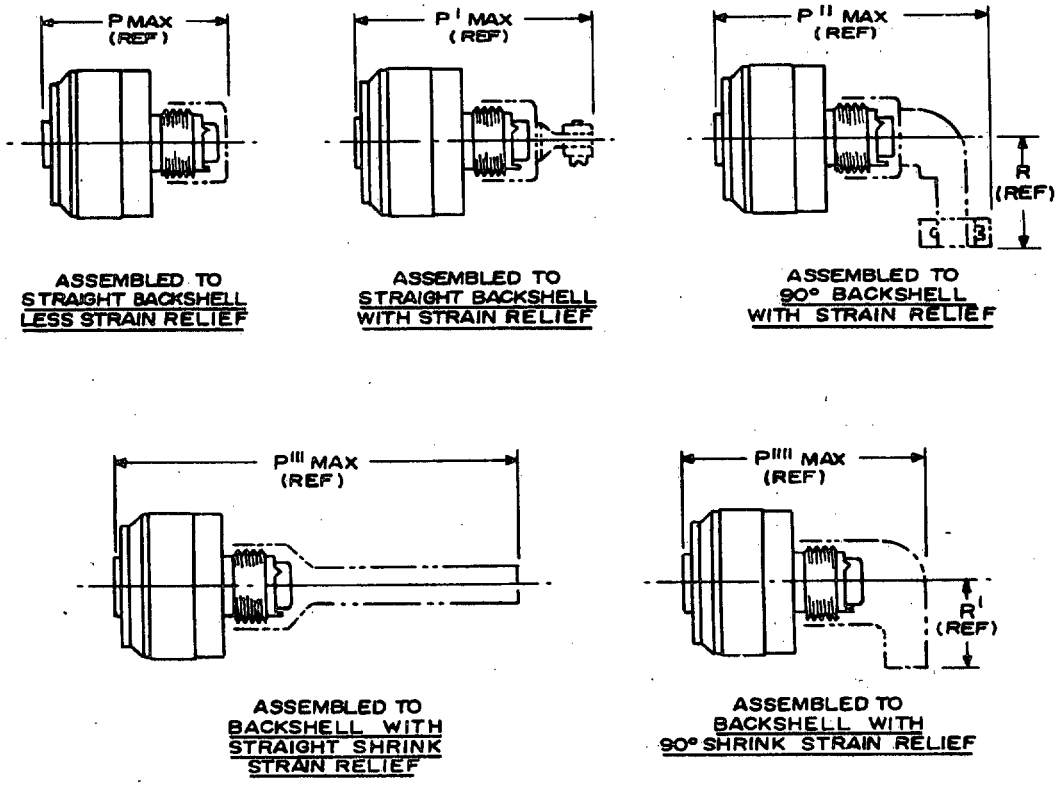


FIGURE 2

TABLE II

SHELL SIZE	POLARIZING POSITION	KEY POSITIONS				INSERT POSITION
		A°	B°	C°	D°	E°
ALL	N	105	140	215	265	0
10,12,14, 16,18,20 22,24 NOTE 8	1	105	140	215	265	10
	2	105	140	215	265	20
	3	105	140	215	265	30
	4	105	140	215	265	40
18	5	105	140	215	265	50
8,10	6	102	132	248	320	0
	7	80	118	230	312	0
	8	35	140	205	275	0
	9	64	155	234	304	0
10	10	25	115	220	270	0
12,14,16 18,20 22,24	6	18	149	192	259	0
	7	92	152	222	342	0
	8	84	152	204	334	0
	9	24	135	199	240	0
	10	98	152	268	338	0

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ASSEMBLED TO STRAIGHT BACKSHELL LESS STRAIN RELIEF

ASSEMBLED TO STRAIGHT BACKSHELL WITH STRAIN RELIEF

ASSEMBLED TO 90° BACKSHELL WITH STRAIN RELIEF

ASSEMBLED TO BACKSHELL WITH STRAIGHT SHRINK STRAIN RELIEF

ASSEMBLED TO BACKSHELL WITH 90° SHRINK STRAIN RELIEF

FIGURE 3

TABLE III

SHELL SIZE	P	P'	P''	P'''	P''''	R	R'
8	1.457	1.869	2.018	5.838	1.848	.830	4.585
10	↑	1.979	2.100	6.528	1.888	.880	5.390
12	|	2.124	2.293	7.348	1.928	.950	6.330
14	|	2.124	2.353	7.348	1.928	1.010	6.330
16	|	2.229	2.502	8.068	2.028	1.070	6.710
18	|	2.474	2.614	8.068	2.028	1.130	6.710
20	|	2.604	2.739	9.068	2.148	1.190	7.740
22	↓	2.729	2.864	9.068	2.148	1.260	7.740
24	1.457	2.854	2.989	9.068	2.228	1.320	8.030

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TABLE IV

Insert Arrangements Requiring Reduced Diameters for Raised Seal Barriers on Outer Row of Contact Cavities.

Shell Size	Insert Arrange-ments	Contact Cavities
8	-3	1, 2 and 3

NOTES:

1. For alternate polarizing positions, the connector insert is rotated in respect to the shell.

2. In "normal position" (position "N") the insert vertical centerline coincides with the center line of the master key of the shell.

3. In "alternate insert positions: (positions 1, 2, 3, 4 and 5) the pin insert is rotated counter clockwise relative to the centerline of the master key of shell.

4. In the "alternate keying position" (positions 6, 7, 8, 9, and 10) the keys are positioned as indicated in Table II with reference to the master key.

5. Four keys (MMC) and insert shall be located within .004 either side of (TP) relative to master key (MMC) and shell O.D. (MMC).

6. Dimensions in inches. Unless otherwise specified, tolerance shall be .XX ± .010 and .XXX ± .005, angular, X° ± 1°0' and X°X' ± 30'.

7. The dimensions showing relationship of coupling ring to shell apply only when coupling ring is in full forward position.

8. Alternate polarizing positions 1, 2, 3, 4, and 5 are for interchangeability use only. Not recommended for new design.

9. The quick-disconnect connector plug is intended to be used with Series III threaded coupling receptacle (MIL-C-83723/82 and /84 with adapter (MIL-C-83723/70) permanently attached to the receptacle.

10. Coupling of the quick-disconnect plug connector is to be accomplished by a straight axial push with no cocking of the coupling mechanism.

11. Uncoupling of the quick-disconnect plug connector is to be accomplisned by a straight axial pull on the outer coupling ring.

REQUIREMENTS:

1. For insert arrangements see MS24260 through MS24263, MS27561, and Supplement 2. Cavity identification shall be per MIL-C-83723.

2. Rear hardware must be used per MIL-C-83723/15 and /16(USAF).

3. Unless otherwise specified in the purchase order:

a. Contacts shall be furnished with the connector. The total quantity snall include 10% or 1, whichever is greater, spare contacts of each size contacts used. See MIL-C-83723/33.

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b. Grommet seal plugs shall be furnished with the connector. Total quantity shall be 15% of each contact size or 3, whichever is greater. See MIL-C-83723/28 (USAF).

c. The insertion/extraction tool for each size contact shall be included in the connector package. See MIL-C-83723/31(USAF) or MS27534(USAF).

4. Connector assembly must include a backshell to meet the moisture sealing requirements. See requirement 2.

5. The coupling mechanism of the connector plug shall withstand a minimum tensile load of 100 lbs. Load to be applied to rear accessory hardware in a straight axial direction.

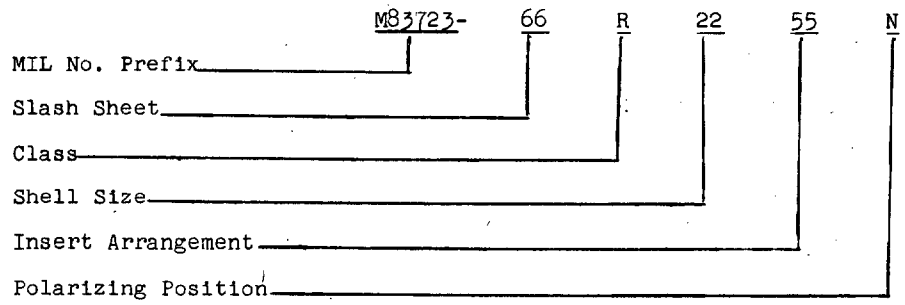
6. The quick-disconnect connector plug shall meet the vibration and shock requirements per paragraph 3.6.19 and 3.6.20 of MIL-C-83723 and the coupling mechanism shall maintain the plug and receptacle in a locked condition.

7. The tensile force required to unlock the coupling mechanism shall be 5 lbs. minimum and 20 lbs. maximum.

8. Locking mechanism as required to meet performance requirements of MIL-C-83723.

9. Material: Class A - Aluminum shell per QQ-A-367; Black Anodize, Non-Conductive Finish per MIL-A-8625.
Class G - Stainless Steel per QQ-S-764.
Class R - Aluminum shell per QQ-A-367.

PART NUMBER EXAMPLE:



Custodians:

Air Force - 85

Review Activities:

Air Force - 11, 70, 80, 84

Preparing Activity:

Air Force - 85

Project No. 5935-F980-1

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-83723/66 (USAF) Connectors, Electrical (Circular, Environment Resisting FSC 5935

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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